

Loopia DNS Wildcard Pitfall

Why Let's Encrypt Certificates Fail with Loopia Domains

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Executive Summary

Loopia (a Swedish domain registrar and hosting provider) pre-configures wildcard DNS records for all registered domains. These wildcard records point to Loopia's hosting servers, which run an ACME responder that automatically answers Let's Encrypt certificate challenges. This causes certificate validation failures when you try to use your domain with external services like Pangolin, Traefik, or any self-hosted infrastructure.

The Problem

Default Loopia DNS Configuration

When you register a domain with Loopia, they automatically create the following DNS records:

Record	Type	Value	Purpose
@	A	194.9.94.85	Root domain → Loopia hosting
@	A	194.9.94.86	Root domain → Loopia hosting (backup)
*	A	194.9.94.85	ALL subdomains → Loopia hosting
*	A	194.9.94.86	ALL subdomains → Loopia hosting (backup)
www	A	194.9.94.85	www subdomain → Loopia hosting

The wildcard record (*) is the critical problem. It means ANY subdomain that doesn't have an explicit A record will resolve to Loopia's servers.

What Loopia's Servers Do

Loopia's hosting servers (194.9.94.85 and 194.9.94.86) run an nginx server with a built-in ACME responder. When any request hits the path `/.well-known/acme-challenge/{token}`, Loopia's server responds with:

```
{token}.{loopia-account-thumbprint}
```

In our case, Loopia's thumbprint was: `-f_daEYxVOFls4aupfol2f4PA8ikqBUw-4tU6dotcK8`

This is a 'helpful' feature for customers using Loopia's web hosting - they can automatically get Let's Encrypt certificates without configuration. However, it's catastrophic for anyone self-hosting.

How ACME HTTP-01 Challenge Works

When you request a Let's Encrypt certificate, the following happens:

1. Your server (Traefik) requests a certificate for pangolin.yourdomain.com
2. Let's Encrypt generates a random token and sends it to your server
3. Your server must respond at `http://pangolin.yourdomain.com/.well-known/acme-challenge/{token}`
4. The response must be: `{token}.{your-account-thumbprint}`
5. Let's Encrypt uses **multiple validation servers worldwide** to verify
6. If ALL validators see the correct response, certificate is issued

The thumbprint is derived from your ACME account's public key. Each account has a unique thumbprint.

What Goes Wrong

Even when your specific subdomain (pangolin.yourdomain.com) correctly points to your VPS:

Scenario:

- Your A record: pangolin.yourdomain.com → 170.75.172.2 (your VPS)
- Wildcard record: *.yourdomain.com → 194.9.94.85 (Loopia)

What happens during validation:

1. Let's Encrypt validator in USA resolves pangolin.yourdomain.com → 170.75.172.2 ✓
2. Let's Encrypt validator in Europe resolves pangolin.yourdomain.com → 170.75.172.2 ✓
3. But DNS caching, geographic variations, or resolver quirks may cause some validators to:
 - Get a cached wildcard response
 - Follow a different resolution path
 - Hit Loopia's servers instead

When a validator hits Loopia's server:

- Expected response: `{token}.{YOUR-thumbprint}`
- Actual response: `{token}.{LOOPIA-thumbprint}`
- Result: **VALIDATION FAILED**

The Error Message

You'll see this in Traefik logs:

```
ERR Unable to obtain ACME certificate for domains
error="The key authorization file from the server did not match this challenge.
Expected "{token}.{your-thumbprint}"
(got "{token}.-f_daEYxV0Fls4aupfol2f4PA8ikqBUw-4tU6dotcK8")
```

The 'got' thumbprint will always be the same (Loopia's) regardless of how many times you retry, delete acme.json, or restart containers. This is the key diagnostic indicator.

How to Diagnose This Issue

Step 1: Check for Wildcard Records

```
# Query the authoritative nameserver directly
dig +short \*.yourdomain.com @ns1.loopia.se

# If this returns 194.9.94.85 or 194.9.94.86, you have the problem
```

Step 2: Test What Loopia Responds With

```
# Hit Loopia's server directly with your domain's Host header
curl -v http://194.9.94.85/.well-known/acme-challenge/test \
-H "Host: pangolin.yourdomain.com"

# If you see a response like:
# test.-f_daEYxV0Fls4aupfol2f4PA8ikqBUw-4tU6dotcK8
# That's Loopia's ACME responder interfering
```

Step 3: Verify Your Own Server Responds Correctly

```
# Hit your VPS directly
curl -v http://YOUR_VPS_IP/.well-known/acme-challenge/test \
-H "Host: pangolin.yourdomain.com"

# Should return 404 Not Found (correct - no active challenge)
```

The Solution

Step 1: Delete the Wildcard DNS Records

In Loopia's DNS editor:

1. Go to your domain's DNS settings
2. Click 'DNS-editor'
3. Find the row with '*' (asterisk)
4. Click 'Ta bort' (Delete) on BOTH A records (194.9.94.85 and 194.9.94.86)
5. If records reappear, delete the entire '*' subdomain entry

Step 2: Verify Deletion at Authoritative Server

```
# Should return empty
dig +short \*.yourdomain.com @ns1.loopia.se
```

Step 3: Wait for DNS Propagation

The wildcard records had TTL 3600 (1 hour). Let's Encrypt validators may cache them. Check global DNS propagation:

```
# Check Google's DNS
```

```
curl -s "https://dns.google/resolve?name=*.yourdomain.com&type=A"

# Status:3 (NXDOMAIN) means the wildcard is gone
```

Step 4: Retry Certificate Generation

```
# Delete old ACME state
sudo rm -f config/letsencrypt/acme.json

# Restart Traefik
docker restart traefik

# Check logs
docker logs traefik 2>&1 | tail -20
```

Prevention for Future Domains

When registering a new domain with Loopia (or any registrar), immediately:

1. Go to DNS settings before doing anything else
2. Delete any wildcard (*) records
3. Delete any default A records pointing to the registrar's IPs
4. Only create the specific A records you need

Why This Is Bad Practice by Loopia

This 'feature' is problematic because:

- It assumes all customers want managed hosting
- It silently interferes with self-hosting setups
- The ACME responder returns valid-looking but wrong responses
- There's no warning during domain registration
- The wildcard can 'snap back' after deleting individual records
- The error message doesn't clearly indicate DNS is the problem

Other Registrars That May Have Similar Issues

While this document focuses on Loopia, other registrars that bundle hosting services may have similar configurations. Always check for wildcard records when:

- Registering a new domain
- Transferring a domain from a hosting provider
- Seeing persistent ACME thumbprint mismatches

Quick Reference

Symptom	Cause	Solution
Same "got" thumbprint every retry	Loopia ACME responder	Delete wildcard DNS
Thumbprint: -f_daEYxV...	Loopia's account key	Delete wildcard DNS
Wildcard reappears after deletion	Loopia auto-recreates	Delete entire * subdomain
Works locally, fails with LE	LE validators hit Loopia	Wait for DNS TTL + retry